

E-Commerce Order Processing & Discount Engine Specification

1. System Overview

The Order Engine manages cart calculations, tiered customer discount rules, tax estimations, and checkout validation. This document specifies the business logic, exception handling, and data models required for implementation.

2. Customer Tier Definitions

Customers belong to one of three tier classifications:

* **STANDARD**: Default tier for all registered users. Receive standard pricing with no automated baseline discounts.

* **SILVER**: Awarded after accumulating \$500 in total lifetime spend. Receives a baseline 5% discount on all eligible items.

* **GOLD**: Awarded after accumulating \$2,000 in total lifetime spend. Receives a baseline 15% discount on all eligible items and qualifies for free priority shipping.

3. Core Business Logic & Pricing Rules

3.1 Item Eligibility and Quantity Constraints

1. Every order item must contain a valid `product\_id` (string), `unit\_price` (float > 0.0), and `quantity` (integer > 0).
2. Maximum quantity per item in a single order is 50 units. If an item exceeds 50 units, throw an `InvalidQuantityException`.
3. Digital products (marked with `is\_digital = True`) are exempt from shipping fees and physical packaging rules.

3.2 Tiered Discount Calculation

Discounts are calculated sequentially on eligible line items:

* **Base Tier Discount**: Applied directly to the item price based on customer status (SILVER = 5%, GOLD = 15%).

* **Bulk Volume Discount**: If the line item quantity is 10 or more, apply an additional 5% bulk discount **after** applying the base tier discount.

* **Maximum Combined Discount Cap**: Under no circumstances may total item-level discounts exceed 25% of the original unit price.

3.3 Tax Calculation Rules

* Standard domestic tax rate is **8.5%** applied to the post-discount subtotal.

* Digital products carry a reduced tax rate of **5.0%**.

* Orders shipped to non-taxable regions (indicated by `tax\_exempt = True` on customer profile) incur **0.0%** tax.

3.4 Shipping Fee Calculation

- * Standard Shipping: Flat fee of ****\$9.99**** for physical orders with subtotal under \$100.00.
- * Free Shipping: Applied automatically if physical order subtotal (post-discount) is \$100.00 or higher, OR if customer tier is ****GOLD****.
- * Express Shipping: Optional add-on of ****\$19.99****, regardless of tier or order total.

4. Required Function Signatures

4.1 `calculate\_order\_subtotal`

Calculates item subtotal applying tier and bulk volume discounts.

****Parameters:****

- * ``items`` (List[Dict]): List of cart item dictionaries containing ``product_id``, ``unit_price``, ``quantity``, and ``is_digital``.
- * ``customer_tier`` (String): Tier indicator (``"STANDARD"`, `"SILVER"`, `"GOLD"`.)`

****Returns:****

- * ``Tuple[float, float]``: (`raw_subtotal`, `discounted_subtotal`)

****Exceptions:****

- * Raises ``InvalidQuantityException`` if ``quantity`` `<= 0`` or ``quantity`` `> 50``.
- * Raises ``InvalidPriceException`` if ``unit_price`` `<= 0.0``.
- * Raises ``UnknownCustomerTierException`` if ``customer_tier`` is not one of the allowed values.

4.2 `calculate\_final\_order\_total`

Computes total order amount including subtotal, tax, and shipping.

****Parameters:****

- * ``discounted_subtotal`` (float): The subtotal returned from ``calculate_order_subtotal``.
- * ``has_physical_items`` (bool): True if at least one item requires physical delivery.
- * ``customer_tier`` (str): Tier classification string.
- * ``tax_exempt`` (bool): Optional flag, defaults to ``False``.
- * ``express_shipping`` (bool): Optional flag, defaults to ``False``.

****Returns:****

- * ``Dict[str, float]``: Structured result containing:
 - * ``"subtotal"`, `"tax"`, `"shipping"`, `"grand_total"`.`

5. Exception Handling Matrix

Error Condition	Exception Class	HTTP Status Code	Error Message
Item quantity zero or negative	<code>InvalidQuantityException</code>	400 Bad Request	"Item quantity must be between 1 and 50."
Item price zero or negative	<code>InvalidPriceException</code>	400 Bad Request	"Unit price must be greater than 0.00."
Invalid customer tier string	<code>UnknownCustomerTierException</code>	422 Unprocessable	"Unrecognized customer tier parameter."
Empty cart submitted	<code>EmptyCartException</code>	400 Bad Request	"Order processing requires at least one item."